

CSCI 160 Homework 4 — Answer Key

1. Simplify $F(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 10)$ with a four-variable K-map, then draw the NAND circuit.

K-map and grouping. Place a 1 in each listed cell and loop the 1's.

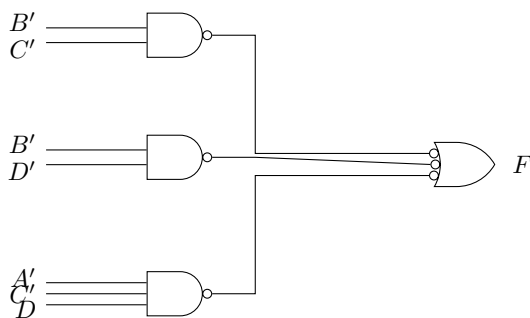
		CD			
		00	01	11	10
AB	00	1	1	0	1
	01	0	1	0	0
	11	0	0	0	0
	10	1	1	0	1

(Red loop: cells 0, 1, 8, 9) $B'C'$
(Blue loop: cells 0, 2, 8, 10) $B'D'$
(Green loop: cells 1, 5) $A'C'D$

The loops are $B'C'$ (cells 0, 1, 8, 9), $B'D'$ (cells 0, 2, 8, 10), and $A'C'D$ (cells 1, 5), so

$$F = B'C' + B'D' + A'C'D$$

NAND circuit. The function is in SOP form, so each AND term is a NAND (AND-invert) gate and the final OR is a NAND (invert-OR) gate. The single literal D branches into both inputs of its gate, as in the lecture example.



2. Simplify $F(A, B, C, D) = \prod M(0, 1, 2, 3, 4, 8, 12)$ with a four-variable K-map, then draw the NOR circuit.

K-map and grouping (loop the 0's). For a POS we place 0's at the listed maxterms and loop the 0's to read off F' .

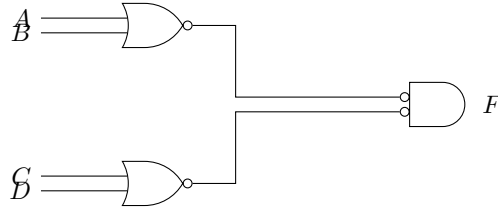
		CD				
		00	01	11	10	
00		0	0	0	0	$A'B'$
01		0	1	1	1	
11	AB	0	1	1	1	
10		0	1	1	1	
		$C'D'$				

The 0-loops are $A'B'$ (cells 0, 1, 2, 3) and $C'D'$ (cells 0, 4, 8, 12), giving the complement $F' = A'B' + C'D'$. Flip it with DeMorgan to get POS form:

$$F = (A'B' + C'D')' = (A + B)(C + D).$$

$$F = (A + B)(C + D)$$

NOR circuit. The function is in POS form, so each OR term is a NOR (OR-invert) gate and the final AND is a NOR (invert-AND) gate.



3. Equality of two 8-bit numbers using XNOR gates.

A single XNOR is a 1-bit comparator: $A_i \odot B_i = 1$ exactly when the two bits agree. Two numbers are equal only when *every* bit position agrees, so AND the eight per-bit XNORs:

$$Equal = \prod_{i=0}^7 (A_i \odot B_i) = (A_7 \odot B_7)(A_6 \odot B_6) \cdots (A_0 \odot B_0).$$

Evaluate for $A = 10110100$ and $B = 10010100$. Compare bit by bit:

position i	7	6	5	4	3	2	1	0
A_i	1	0	1	1	0	1	0	0
B_i	1	0	0	1	0	1	0	0
$A_i \odot B_i$	1	1	0	1	1	1	1	1

Bit position 5 differs (1 vs. 0), so $A_5 \odot B_5 = 0$. The AND of all eight terms is therefore

$$Equal = 1 \cdot 1 \cdot 0 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 0.$$

$$\boxed{Equal = 0 \quad \Rightarrow \quad A \neq B}$$

4. Even-parity bit for the 4-bit message $wxyz = 1011$.

With even parity, P is chosen so the whole word $wxyzP$ has an even number of 1's. Equivalently $P = w \oplus x \oplus y \oplus z$ (the odd function): $P = 1$ when the message has an odd number of 1's, $P = 0$ when even.

The message 1011 has three 1's — an *odd* count — so P must be 1 to make the total even:

$$P = w \oplus x \oplus y \oplus z = 1 \oplus 0 \oplus 1 \oplus 1 = 1.$$

The transmitted word is

$$\boxed{wxyzP = 10111}$$

which has four 1's — even, as required.